

Practical Project Assignment

Learning Goals

Like in other knowledge application domains, understanding how to analyse and interpret data is crucial for making informed decisions in different dimensions of software (SW) engineering, from software development to project management. Data Analysis (DA) is a powerful technique that allows SW practitioners to summarise the main characteristics of a dataset, often using visual methods to report results. By conducting DA, software engineers can gain insights into the data and identify patterns, trends, or anomalies.

This assignment aims to introduce students to the concept of data analysis, its project lifecycle, and underlying methodologies. Students will be exposed to various problems and will use DA techniques to analyse real-world datasets (for instance, related to software development processes or other domains). The expected learning outcomes are as follows.

1. Understand how selected machine learning (ML) algorithms work in the context of data analysis, both empirically and theoretically.
2. Understand how the evaluation and benchmarking of ML algorithms is carried out.
3. Understand the practical application of ML algorithms to support the solution of data analysis problems.
4. Develop critical-thinking skills in data analysis.

Assignment description

The assignment consists of the following activities:

1. **Data Collection:** Students will select a real-world dataset of their choice;
2. **Data Preprocessing:** Before conducting DA tasks, students must preprocess the data. This may involve handling missing values, converting data types, or normalising the data;
3. **Data Analysis:** Students will perform exploratory DA on the pre-processed dataset using appropriate techniques. This may include:
 - a. **Descriptive Statistics:** Compute summary statistics (e.g., mean, median, standard deviation) for each metric.
 - b. **Data Visualization:** Create visualisations (e.g., histograms, box plots, scatter plots) to explore the distribution and relationships between different metrics.

- c. Correlation Analysis: Analyse the correlation between different metrics to identify potential relationships.
 - d. ML tasks: selection and application of different algorithms (supervised and unsupervised) and their appropriate evaluation.
4. Insights and Interpretation: Based on the results of the DA, students will interpret their findings and draw conclusions. They may identify trends, outliers, or patterns in the data and discuss their implications for the problem at hand.
5. Report and Presentation: Students will prepare a report summarising their DA findings and insights, and will make a presentation of their main results.

Schedule, deadlines, and deliverables

Week 15 (2026-06-08): assignment submissions

- code/notebook (commented), datasets, updated slide decks

Week 15 (2026-06-11): final presentations (in-class presentation)

- Oral presentation with slide deck support (Moodle submission)

Groups

The practical assignment is mandatory and should be performed by groups of four students. You should sign up for a group on Moodle by enrolling in one of the groups available. Please make sure to follow the enrolment criteria below:

- Students in class 1MESW01 should choose a group from WG1_01 to WG1_07.
- Students in class 1MESW02 should choose a group from WG2_01 to WG2_07.

Grading criteria

- Problem description, understanding, and clarity (10%)
- Exploratory data analysis (20%)
- Empirical data analysis study (20%)
- Comprehension of the selected algorithms (10%)
 - Handling of data characteristics (e.g. unbalanced, missing values)
- Main findings and conclusions (10%)
- Notebook/code (10%)
 - organisation, implementation/code, and comments
- Presentation - mandatory
 - slides (10%)
 - presentation and discussion (10%)